Vidyamandir Classes
Gurukul for IITJEE & Medical Preparation

Illuminati - 2019

Advanced Physics Assignment -3B

Electrostatics, DC-Circuits, Capacitors, Magnetism, EMI, AC-Circuits

Errata for questions

18. This time, the deviations of the needles are θ_1' and θ_2' respectively. Find $r2$.

22. The resistance of each voltmeter is infinite

(B) Reading of ammeter A_1 is zero and A_2 is 1/25 A

In diagram : 90Ω resistance should be 900Ω

24. $C = \frac{20}{\pi}\mu F$

29. **(D)** The field at points on z-axis which are on either side of origin inside the sphere are in opposite direction

SECTION-1	SINGLE CORRECT ANSWER TYPE

1.(C) The acceleration of centre of mass of system of particles is

$$a_{cm} = \frac{(q_1+q_2)}{2m}E$$

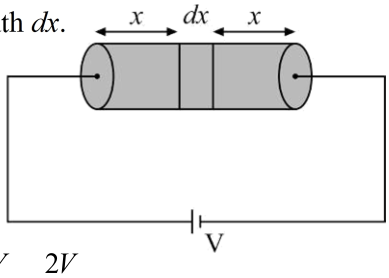
$\therefore$ x-coordinate of centre of mass at $t=2$ second is

$$x_{cm} = \frac{1}{2}a_{cm}t^2 = \frac{1}{2}\frac{(q_1+q_2)}{2m}E\times 2^2 = \frac{q_1+q_2}{m}E$$

Let the x-coordinates of q_1 and q_2 at $t=2$ sec be x_1 and x_2; $[x_1 = 2a$ at $t=2$ sec$]$

$$\therefore \quad x_{cm} = \frac{mx_1+mx_2}{2m} = \frac{x_1+x_2}{2} \qquad \text{or} \qquad x_2 = 2x_{cm} - x_1 = 2\frac{(q_1+q_2)}{m}E - 2a$$

2.(A) Consider an elemental part of solid at a distance x from left end of width dx.



$$dR = \frac{\rho dx}{\pi a^2} = \frac{\rho_0 x\, dx}{\pi a^2} \quad ; \; R = \int dR = \int_0^L \frac{\rho_0 x\, dx}{\pi a^2} = \frac{\rho_0 L^2}{2\pi a^2}$$

Current through cylinder is $I = \frac{V}{R} = \frac{V\times 2\pi a^2}{\rho_0 L^2}$

Potential drop across element is, $dV = IdR = \frac{2V}{L^2}x\,dx \Rightarrow E(x) = \frac{dV}{dx} = \frac{2V}{L^2}x$

3.(B) Force between plates $F = \frac{Q^2}{2A\epsilon_0} = \frac{\left(\frac{\epsilon_0 A}{x}V\right)^2}{2A\epsilon_0} = \frac{\epsilon_0 AV^2}{2x^2}$ where x is separation between plates

$dW = F\,dx$

$$W = \int_d^{2d} \frac{\in_0 AV^2}{2x^2} dx = \left[-\frac{\in_0 AV^2}{2x} \right]_d^{2d} = \frac{CV^2}{4} = 200\,\mu J$$

Alternate Method

$$U_i + W_B + W_{ext} = U_f + loss$$

Process is slow so energy loss is zero, work done by battery $= W_B = QE$

$$Q = Q_f - Q_i = 20 - 40 = -20$$

$$W_B = -20 \times 20 \quad ; \qquad \frac{1}{2} 2 \times 20^2 - 20 \times 20 + W_{ext} = \frac{1}{2} 1 \times 20^2 + 0$$

$$W_{ext} = 200\,\mu J$$

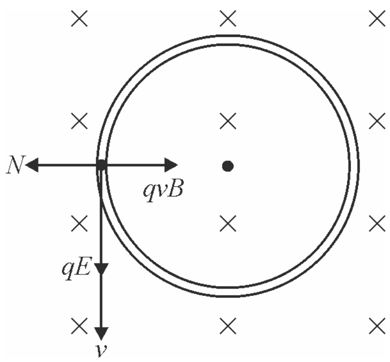
4.(A) Unit vector along z direction is given as $\hat{\mu} = \hat{k}$

Magnetic induction in region is given as $\vec{B} = 3\hat{j} + 4\hat{i}$

Torque vector on the ring acts along the direction $\hat{\mu} \times \vec{B} = 3\hat{j} - 4\hat{i}$

Thus the ring will topple about a point located on the line perpendicular to the direction of torque vector which is $3\hat{i} + 4\hat{j}$. Thus the point on ring located in this direction is (3, 4).

5.(B) Induced electric field at a distance R from centre is given as



$$E = \frac{R}{2}\frac{dB}{dt} = \frac{R}{2}\alpha$$

Speed attained by the bead is given as

$$v = \frac{qE}{m} t$$

For circular motion of bead, from below figure, we have

$$qvB - N = \frac{mv^2}{R} \qquad \Rightarrow \qquad N = \frac{\alpha q^2 Rt}{4m}(2B_0 + \alpha t)$$

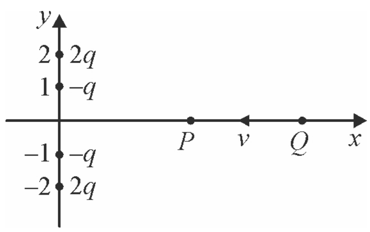
6.(A) Since, current leads emf, therefore this is an R-C circuit.

$$\therefore \qquad \tan\phi = \frac{X_C - X_L}{R} \qquad (\because \phi = 45°)$$

$\Rightarrow \qquad X_C = R \qquad$ ($X_L = 0$ as there is no inductor) ;

$$\frac{1}{\omega C} = R \qquad \Rightarrow \qquad \frac{1}{100 \times 10 \times 10^{-6}} = R \qquad \Rightarrow \qquad R = 10^3\,\Omega$$

7.(B) There exists a point *P* on the *x*-axis (other than the origin), where net electric field is zero. Once the charge *Q* reaches point *P*, attractive forces of the two negative charge will dominate and automatically cause the charge *Q* to cross the origin.



Now if *Q* is projected with just enough velocity to reach *P*, its K.E. at *P* is zero.

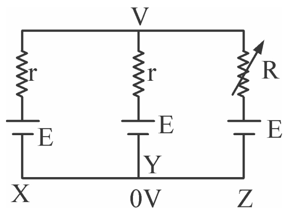
But while being attracted towards origin it acquires K.E. & hence its net energy at the origin is positive.

(P.E. at origin = zero).

8.(B) $V = \dfrac{K\,\vec{p}.\vec{r}}{r^3} = \dfrac{9 \times 10^9 \times (2\hat{i} - 3\hat{j} + 4\hat{k}).(2\hat{i} + 2\hat{j} - \hat{k})}{[2^2 + 2^2 + 1^2]^{3/2}} = \dfrac{9 \times 10^9 \times (4 - 6 - 4)}{27}$

$$V = -2 \times 10^9 \text{ volts}$$

9.(C) Redrawing the given circuit diagram as shown below :



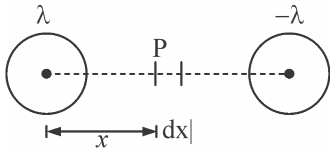
Using point potential theory

$$\frac{V-E}{r}+\frac{V-E}{r}+\frac{V-E}{R}=0 \quad \Rightarrow \quad (V-E)\left(\frac{2}{r}+\frac{1}{R}\right)=0$$

As $\frac{2}{r}+\frac{1}{R}\neq 0$ so $V-E=0$

So, current through $R, i=\frac{V-E}{R}=0$ whatever be the value of R.

10.(C) Let us give equal and opposite charges to two wires so that they would have linear charge density as $+\lambda$ and $-\lambda$



Electric field at point P.

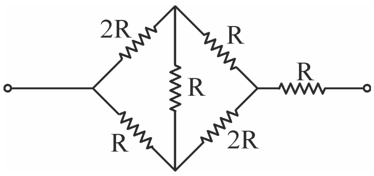
$$E=\frac{\lambda}{2\pi\in_0 x}+\frac{\lambda}{2\pi\in_0(\eta a-x)} \qquad \int dV=-\int E\,dx=-\int_a^{\eta a-a} E\,dx$$

where a is radius of wire $\Rightarrow C=\frac{\lambda}{|V|}=\frac{\pi\in_0}{\ln\eta}$

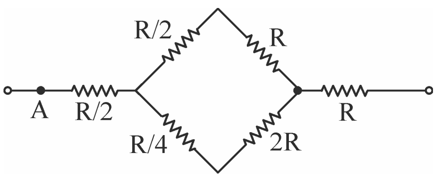
11.(C) Let σ be the charge density of conducting plate and V be the volume of either dielectric

$$\therefore \quad \frac{U_1}{U_2}=\frac{\left(\frac{1}{2}K_1\in_0 E_1^2\right)V}{\left(\frac{1}{2}K_2\in_0 E_2^2\right)V}=\frac{K_1}{K_2}\frac{\left(\frac{\sigma}{K_1\in_0}\right)^2}{\left(\frac{\sigma}{K_2\in_0}\right)^2}=\frac{K_2}{K_1}$$

12.(D)



From delta – star transformation, the circuit can be redrawn as

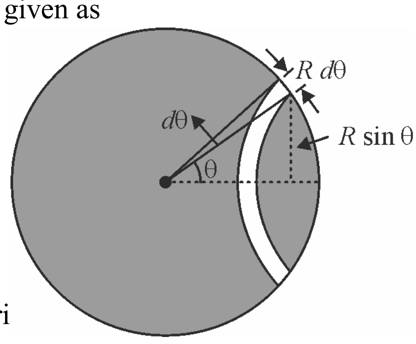


$$R_{eq}=\frac{\frac{3R}{2}\times\frac{9R}{4}}{\frac{3R}{2}+\frac{9R}{4}}+R+\frac{R}{2}=\frac{12}{5}R$$

$$I=\frac{V}{2R}\,; \qquad I'=\frac{5V}{12R} \Rightarrow \frac{I'}{I}=\frac{5}{6}$$

13.(A) Figure shows the charged sphere on which we consider an elemental ring at an angle θ from the axis of angular width $d\theta$. The surface area of the elemental ring is given as

$dS=2\pi R\sin\theta\times Rd\theta$



The charge on the elemental ring surface is given as

$dq=\sigma dS$

Current due to rotating charge on the elemental ring is given as

$$di=\frac{dq\omega}{2\pi} \Rightarrow di=\frac{(2\pi\sigma R^2\sin\theta d\theta)\omega}{2\pi} \Rightarrow di=\sigma\omega R^2\sin\theta d\theta$$

Magnetic induction at the centre O due to the current in elemental ri

$$dB=\frac{\mu_0 di(R\sin\theta)^2}{2(R^2\cos^2\theta+R^2\sin^2\theta)^{3/2}} \Rightarrow dB=\frac{\mu_0 di\sin^2\theta}{2R}$$

Substituting the value of current di we get

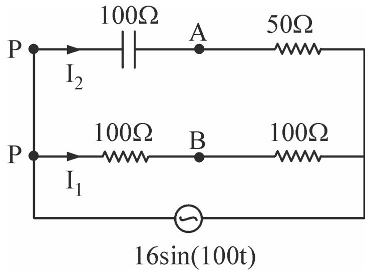
$$dB = \frac{\mu_0}{2} \times \sigma\omega R^2 \sin\theta d\theta \times \frac{\sin^2\theta}{R} \quad \Rightarrow \quad dB = \frac{1}{2}\mu_0\sigma\omega R \sin^3\theta d\theta$$

Net magnetic induction at O is given by integrating above expression within limits of θ from 0 to π given as

$$B = \int dB = \frac{1}{2}\mu_0\sigma\omega R \int_0^{\pi} \sin^3\theta d\theta \quad \Rightarrow \quad B = \frac{1}{2}\mu_0\sigma\omega R\left[\frac{1}{3}\cos^3\theta - \cos\theta\right]_0^{\pi}$$

$$\Rightarrow \quad B = \frac{1}{2}\mu_0\sigma\omega R\left[\frac{1}{3}(-1) - (-1) - \frac{1}{3}(1) + (1)\right]_0^3 \quad \Rightarrow \quad B = \frac{2}{3}\mu_0\sigma\omega R$$

14.(D) $X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-16}} = 100\Omega$



$I_1)_{max} = \frac{16}{200}$

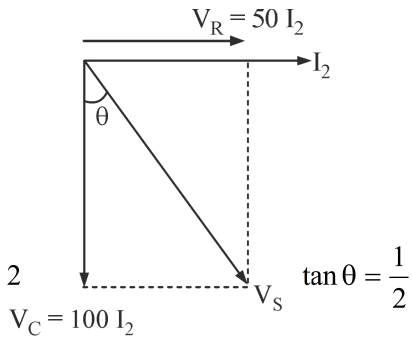
$(V_P - V_B)_{max} = \frac{16}{200} \times 100 = 8$

Also I_1 is in phase with the source voltage and so is $(V_P - V_B)$

$\therefore \quad V_P - V_B = 8\sin 100t \quad \ldots . (1)$

$I_2)_{max} = \frac{16}{\sqrt{100^2 + 50^2}} = \frac{16}{50\sqrt{5}}$

$(V_P - V_A)_{max} = \frac{16}{50\sqrt{5}} \times 100 = \frac{32}{\sqrt{5}}$



From phasor diagram, $(V_P - V_A)$ lags source voltage by $\theta = \tan^{-1} 2$ $\quad \tan\theta = \frac{1}{2}$

$\therefore \quad V_P - V_A = \frac{32}{\sqrt{5}}\sin(100t - \theta) \quad \ldots . (2)$

(1) - (2) :

$V_A - V_B = 8\sin 100t - \frac{32}{\sqrt{5}}\sin(100t - \theta)$

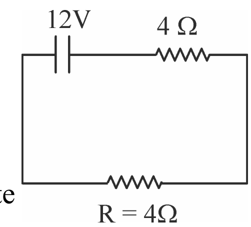
To find $(V_A - V_B)_{max}$, the RHS can be treated as subtraction of two vectors with angle then as θ

$\therefore \quad (V_A - V_B)_{max} = \sqrt{8^2 + \left(\frac{32}{\sqrt{5}}\right)^2 - 2(8)\left(\frac{32}{\sqrt{5}}\right)\cos\theta}$

Putting $\cos\theta = \frac{2}{\sqrt{5}}, (V_A - V_B)_{max} = 8$

15.(C) At resonance

$Z = R$



So, $\quad (I_{rms})_{max} = \frac{V_{rms}}{R} \quad ; \quad 6 = \frac{24}{R}$

$R = 4\Omega$ (Resistance of the coil)

When coil is connected to battery, then current in steady state

$I = \frac{12}{4+4} = 1.5A$

16.(B) Just before S_1 is closed the potential difference across capacitor 2 is $2E$.
Just after S_1 is closed the potential difference across capacitors 1 and 2 are 0 and $2E$ respectively.
Applying KVL to loop $ABCD$ immediately after S_1 is closed.

$$E = -iR_1 + 0 + 2E \qquad \text{or} \qquad i = \frac{E}{R_1} \text{ towards left}$$

17.(D) When DC is applied across a solenoid, then Resistance, i.e. $R = \dfrac{V}{I}$

$$R = \frac{100}{1} = 100\Omega$$

When AC is applied, then $\qquad z = \dfrac{V}{I}$

$$Z = \frac{100}{0.5} = 200\Omega$$

Net impedance,

$$Z = \sqrt{R^2 + X_L^2}$$

$$200 = \sqrt{(100)^2 + X_L^2}$$

$$\Rightarrow \quad (200)^2 = (100)^2 + X_L^2 \quad \Rightarrow \quad 30000 = X_L^2 \quad \Rightarrow \quad X_L = 173.205\Omega$$

Now, $\quad X_L = \omega L$

$$\Rightarrow \quad 173.205 = 2\pi f L \quad \Rightarrow \quad \frac{173.205}{2\pi \times f} = L \quad \Rightarrow \quad \frac{173.205}{2 \times 3.14 \times 50} L$$

$$\Rightarrow \quad 0.55\ \Omega = L \quad ; \quad L = 0.55\Omega$$

18.(C) $I = \dfrac{V}{r_1 + r_2} = k_1\theta_1 = k_2\theta_2 \Rightarrow \dfrac{k_1}{k_2} = \dfrac{\theta_2}{\theta_1}$

$$I_1' = \frac{V}{r_1} = k_1\theta_1',\ I_2' = \frac{V}{r_2} = k_2\theta_2' \quad \Rightarrow \quad \frac{r_2}{r_1} = \frac{k_1\theta_1'}{k_2\theta_2'} = \frac{\theta_2}{\theta_1}\frac{\theta_1'}{\theta_2'}$$

SECTION-2	**MULTIPLE CORRECT ANSWER TYPE**

19.(ACD) Due to sheet the magnetic induction is given as

$$B = \frac{1}{2}\mu_0 k = \frac{\mu_0(2bJ)}{2} = \mu_0 JB$$

The magnetic induction inside the sheet is given as $B = \mu_0 Jx$

20.(AC) Time period of motion is given as

$$T = \frac{2\pi m}{Bq} = \frac{2\pi}{\alpha B_0} \quad ; \quad \text{At} \quad t = \frac{\pi}{\alpha B_0} = \frac{T}{2}$$

Velocity of particle is $-v_0\hat{i} + v_0\hat{k}$

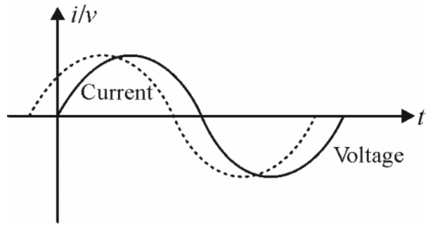
Speed of charge in magnetic field always remains constant so it will be

$$v = v_0\sqrt{2} \quad ; \quad \text{At} \quad t = \frac{2\pi}{\alpha B_0} = T$$

Displacement is equal to pitch which is given as

$$\Delta x = V_0 T = \frac{2\pi V_0}{\alpha B_0} \quad ; \quad \text{At} \quad t = \frac{2\pi}{\alpha B_0} = T$$

Distance is given as $\qquad \Delta x = v \times T = \dfrac{2\sqrt{2}V_0\pi}{\alpha B_0}$



21.(BCD) The graph below shows $V \& I$ phasors.

Current leads the voltage by $\pi / 3$.

Power is positive if $V \& I$ are of same sign

If $\omega \uparrow \Rightarrow \frac{1}{\omega C} \downarrow$ thus angle decreases.

22.(BC) In steady state current through branch of capacitor is zero. Thus current flows through 200Ω and 900Ω and ammeter A_2 in series combination.

Potential difference across the capacitor is given as

$$V_C = \frac{q}{C} = \frac{4 \times 10^{-3}}{100 \times 10^{-6}} = 40V$$

This is also potential drop across 900Ω resistance and 100Ω ammeter A_2 in series of which the total resistance is 1000Ω. This 1000Ω and 200Ω resistance are in series so we use

$$V_2 = V_{200\Omega} = \frac{V_{1000\Omega}}{5} = \frac{40}{5} = 8V$$

Thus battery EMF is given as

$$E = V_{1000\Omega} + V_{200\Omega} = 48V$$

Current through the battery is given as

$$i = \frac{E}{R} \quad \Rightarrow \quad i = \frac{48}{1200} = \frac{1}{25}A$$

Thus options (B) and (C) is correct.

23.(AD) Magnetic field due to outer loop is into the plane within the loop. So $I(\overrightarrow{d\ell} \times \vec{B})$ is radially outward for inner loop.

24.(BD) $$X_C = \frac{1}{100\pi \times \frac{20}{\pi} \times 10^{-6}} = 500\Omega$$

$$X_L = 100\pi \times \frac{5}{\pi} = 500\Omega$$

As $X_C = X_L \quad \Rightarrow$ Voltage across the combination is zero.

So, the resistor is short circulated and as $X_L(30\Omega) > X_C(20\Omega)$

$\therefore$ Circuit is purely inductive.

25.(ABD) Increasing the accelerating voltage means increasing speed of the electron, thereby decreasing time spent between the plates. It will reduce X.

Increasing deflecting voltage means increasing electric field between the plates, making acceleration of electron greater. Increasing distance once again will change electric field between the plates.

26.(ACD) Let the capacitance before insertion of dielectric be C and the resistance be R.

$\because$ $q = q_0 e^{-\frac{t}{RC}}$ and $i = \frac{q/C}{R} = \frac{q}{RC}$

$\because$ Just after insertion of dielectric the capacitance increases.

$\therefore$ The charge just after insertion of dielectric remains same, but the current decreases.

27.(AC) We have $X_L = \omega L = 100 \times 0.5 = 50\Omega$, $X_C = \frac{1}{\omega C} = \frac{1}{100 \times 100 \times 10^{-6}} = 100\Omega$

For branch counting capacitor and resistor :

$Z_1 = 100\sqrt{2}$, $\quad I_1 = \frac{20}{100\sqrt{2}} = \frac{1}{5\sqrt{2}}$

Voltage across $100\Omega = \frac{1}{5\sqrt{2}} \times 100 = \frac{20}{\sqrt{2}} = 10\sqrt{2}$

Phase difference between I_1 and V

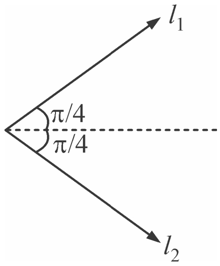
$\cos\phi_1 = \frac{R_1}{Z_1} = \frac{100}{100\sqrt{2}} = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \phi = \frac{\pi}{4}$; $\quad I_1$ leads V by $\frac{\pi}{4}$

For branch containing inductor and resistor :

$Z_2 = 50\sqrt{2}, I_2 = \frac{20}{50\sqrt{2}} = \frac{2}{5\sqrt{2}}$

Voltage access $50\Omega = \frac{2}{5\sqrt{2}} \times 50 = 10\sqrt{2}$

$\cos\phi_2 = \frac{R_2}{z_2} = \frac{50}{50\sqrt{2}} = \frac{1}{\sqrt{2}} \quad \Rightarrow \quad \phi_2 = \frac{\pi}{4}$; I_2 *lags* V by $\frac{\pi}{4}$

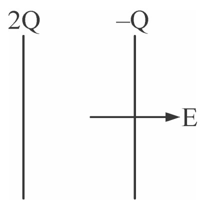


$I_{net} = \sqrt{I_1^2 + I_2^2}$

$I = \sqrt{\frac{1}{25\times 2} + \frac{4}{25\times 2}}$

$= \sqrt{\frac{5}{50}} = \frac{1}{\sqrt{10}} = 0.316$

28.(ABC) $\quad E = \frac{2Q}{2A\in_0} + \frac{Q}{2A\in_0} \quad \Rightarrow \quad E = \frac{3Q}{2A\in_0}$



$E = \frac{3}{2}\frac{Q}{Cd} \quad \Rightarrow \quad Ed = \frac{3Q}{2C} = V$

$F = \left(\frac{2Q}{2A\in_0}\right) \times \frac{(-Q)}{1} = -\frac{Q^2}{A\in_0}$; $\quad F = \frac{Q^2}{A\in_0}$

Energy $= \frac{1}{2}\in_0 E^2 Ad \quad = \frac{1}{2}\in_0 \left(\frac{3Q}{2Cd}\right)^2 Ad = \frac{9}{8}\frac{Q^2}{C}$

29.(AB) Field due to both hemispherical shells is directed towards negative z-axis within the sphere.

30.(AD) While revolving in conical pendulum force equations on the charge can be written as

$$T\sin\theta + F = m\omega^2 (L\sin\theta)$$

$$T\sin\theta = mg \quad \Rightarrow \quad \frac{mg}{\cos\theta}\sin\theta + \theta(\omega L\sin\theta)B = m\omega^2 L\sin\theta$$

$$\Rightarrow \quad B = \frac{m\omega^2 L\sin\theta}{(\omega L\sin\theta)q} - \frac{\frac{mg\sin\theta}{\cos\theta}}{(\omega L\sin\theta)q} \quad \Rightarrow \quad B = \frac{m\omega}{q} - \frac{mg}{q\omega L\cos\theta} = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L\cos\theta}\right]$$

SECTION-3	LINK-COMPREHENSION TYPE

31.(D) $eE = \frac{mv^2}{r} \quad \Rightarrow \quad v = \sqrt{\frac{eE_0 b}{m}}$; $\quad a_c = \frac{eE}{m} = \frac{e}{m}E_0\left(\frac{b}{r}\right)$

32.(C) If $v = \sqrt{\frac{2eE_0 b}{m}}$, centrifugal force > electric force (centripetal force) in the reference frame of proton.

$$\frac{1}{2}m_e v^2 = eE_0 b\int_a^b \frac{dr}{r} \quad \Rightarrow \quad v = \sqrt{\frac{2eE_0 b}{m_e}\ell n\left(\frac{b}{a}\right)}$$

33.(C) v is independent of r.

34.(B) $i_2 = i_3 = I_b$

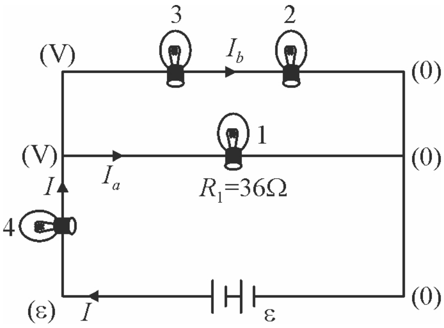
$V_1 = (V_2 + V_3) = V$ and $P_2 = P_3$

$\Rightarrow \quad R_2 = R_3$ and $V_2 = V_3 = V/2$

$P_1 = P_2 = P_3 = P_4$ (given)

$P_1 = \frac{V^2}{36}; \; P_3 = \frac{(V/2)^2}{R_3}$

As $P_1 = P_3 \Rightarrow R_3 = 9\Omega$



35.(A) $I_a = \frac{I}{3}$ and $I_4 = I$

$P_1 = P_4 = 4W$

$\left(\frac{I}{3}\right)^2 R_1 = (I)^2 R_4 = 4W$

$\Rightarrow \quad \frac{R_1}{9} = R_4 \quad \Rightarrow \quad 4\Omega = R_4$

36.(B) Also $I = 1A$

$R_{eq} = 16\Omega$ and $I = 1A$

$\varepsilon = 16V$

37.(C) **38.(D)** **39.(C)**

For t = 0 to $t_o = RC$ seconds, the circuit is of charging type. The charging equation for this time is

$$q = CE\left(1 - e^{-\frac{t}{RC}}\right)$$

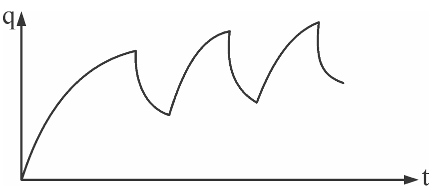
Therefore the charge on capacitor at time $t_o = RC$ is $q_o = CE\left(1 - \frac{1}{e}\right)$

For t = RC to t = 2RC seconds, the circuit is of discharging type. The charge and current equation for this time are

$$q = q_0 e^{-\frac{t-t_0}{RC}} \text{ and } i = \frac{q_0}{RC} e^{-\frac{t-t_0}{RC}}$$

Hence charge at t = 2 RC and current at t = 1.5 RC are

$$q = q_0 e^{-\frac{2RC-RC}{RC}} = \frac{q_0}{e} = \frac{1}{e} CE\left(1 - \frac{1}{e}\right)$$



and $\quad i = \frac{q_0}{RC} e^{-\frac{1.5RC-RC}{RC}} = \frac{q_0}{\sqrt{e}RC} = \frac{E}{\sqrt{e}R}\left(1 - \frac{1}{e}\right)$ respectively

Since the capacitor gets more charged up from t = 2RC to t = 3RC than in the interval t = 0 to t = RC, the graph representing the charge variation is as shown in figure.

40.(B) The force per unit length f between two indefinite parallel straight wires separated by a distance h is.

$$f = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{h}$$

for $I_1 = I_2 = I$ and length $2\pi a$, the force F induced on C_2 by the neighbor coils C_1 is

$$\boxed{F = \frac{\mu_0 a}{h} I^2}$$

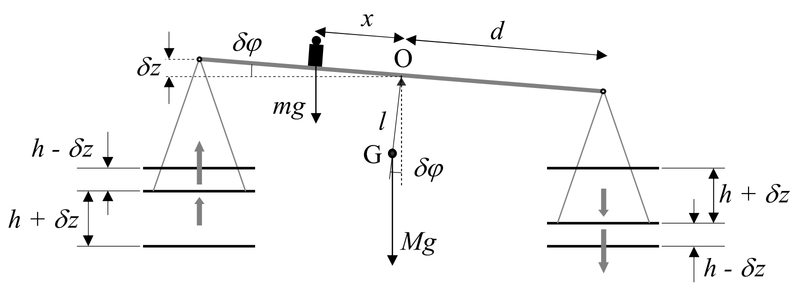
41.(A) In equilibrium $m g x = 4 F d$

Then $m g x = \dfrac{4\mu_0 a d}{h} I^2$ (1)

so that $\boxed{I = \left(\dfrac{m g h x}{4\mu_0 a d}\right)^{1/2}}$

42.(B) The balance comes back towards the equilibrium position for a little angular deviation $\delta\varphi$ if the gravity torques with respect to the fulcrum O are greater than the magnetic torques.

$$Mg\, l sin\delta\varphi + m g x \ \ cos\delta\varphi > 2\mu_0 \, a I^2 \left(\frac{1}{h-\delta z} + \frac{1}{h+\delta z}\right) d \ \ cos\delta\varphi$$



Therefore, using the suggested approximation

$$Mg\, l \ \ sin\delta\varphi + m g x \ \ cos\delta\varphi > \frac{4\mu_0 \, a d I^2}{h}\left(1 + \frac{\delta z^2}{h^2}\right) cos\delta\varphi$$

Taking into account the equilibrium condition (1), one obtains

$$M g l \ \ \sin\delta\varphi > m \ \ g \ \ x \ \ \frac{\delta z^2}{h^2} \ \ \cos\delta\varphi$$

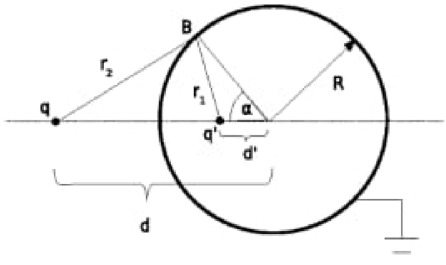
Finally, for $tan \ \ \delta\varphi \approx sin \ \ \delta\varphi = \dfrac{\delta z}{d}$

$$\delta z < \frac{M l h^2}{m x d} \quad \Rightarrow \quad \boxed{\delta z_{max} = \frac{M l h^2}{m x d}}$$

43.(C) **44.(B)** **45.(A)**

Sphere is grounded, its potential vanishes, $V = 0$.

Let us consider an arbitrary point B on the surface of the sphere as depicted in figure.



The distance of point B from the charge q' is

$$r_1 = \sqrt{R^2 + d'^2 - 2Rd'\cos\alpha} \qquad \ldots.(1)$$

Whereas the distance of the point B from the charge q is given with the expression

$$r_2 = \sqrt{R^2 + d^2 - 2Rd\cos\alpha} \qquad \ldots.(2)$$

The electric potential at the point B is

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_2} + \frac{q'}{r_1}\right)$$

This potential must vanish, i.e. its numerical value is 0 V.

$$\frac{q}{r_2} + \frac{q'}{r_1} = 0 \qquad \ldots.(3)$$

Combining (1), (2) and (3) we obtain

$$R^2 + d^2 - 2Rd\cos\alpha = \left(\frac{q}{q'}\right)^2 (R^2 + d'^2 - 2Rd'\cos\alpha)$$

As the surface of the sphere must be equipotential, this condition must be satisfied for every angle α what leads to the following results

$$d^2 + R^2 = \left(\frac{q}{q'}\right)^2 (R^2 + d'^2) \qquad \text{and} \qquad dR = \left(\frac{q}{q'}\right)^2 (d'R)$$

By solving, we obtain the expression for the distance d' of the charge q' from the center of the sphere

$d' = \frac{R^2}{d}$ and the size of the charge q' ; $q' = -q\frac{R}{d}$

Finally, the magnitude of force acting on the charge q is

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q^2Rd}{(d^2 - R^2)^2}$$

The force is apparently attractive.

SECTION-4	COLUMN MATCHING TYPE

46. **A-**r; **B-**q, t; **C-**t; **D-**s

$\vec{F} = q\vec{E} + q\left(\vec{V} \times \vec{B}\right)$

If $\vec{u} = 0,\ \vec{B} = B_x\hat{i}$ and $\vec{E} = E_y\hat{j}$

then charge particle will start to move in y-direction due to electric field and as it acquires velocity it will experience force due to magnetic field and will move in a cycloid path. Similarly, one can find path for other cases.

(A) If $B_y = B_z = E_x = E_z = 0$; $u = 0$ then path is cycloid.

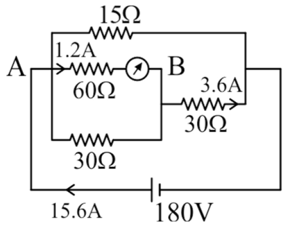
(B) If $E = 0,\ u_xB_x + u_yB_y \neq -u_zB_z$ then path is helix with uniform pitch and constant radius or straight line.

(C) If $\vec{u} \times \vec{B},\ \vec{u} \times \vec{E} = 0$ then path is straight line.

(D) If $\vec{u} \perp \vec{B}$, $\vec{B} \parallel \vec{E}$ then path is helix with variable pitch and constant radius.

47. **A**-p, s, t ; **B**-q, r ; **C**-q, t ; **D**-p

When switch S_1 is closed



Similarly analyse for other parts.

48. **A**-p, s; **B**-q, s; **C**-q, s; **D**-s

(A) Electrostatic potential energy $= \frac{1}{4\pi\in_0}\frac{(-Q)^2}{2a} = \frac{Q^2}{8\pi\in_0 a}$

(B) Electrostatic potential energy $= \frac{1}{4\pi\in_0}\left[\frac{(-Q)\times(-Q)}{5a/2} + \frac{(-Q)^2}{2(5a/2)}\right] = \frac{3}{20}\frac{Q^2}{\pi\in_0 a}$

(C) Electrostatic potential energy $= \frac{1}{4\pi\in_0}\frac{3Q^2}{5a} = \frac{3}{20}\frac{Q^2}{\pi\in_0 a}$

(D) Electrostatic potential energy $= \frac{1}{4\pi\in_0}\left[\frac{3Q^2}{5a} + \frac{(-Q)^2}{2(2a)} + \frac{(-Q)\times(-Q)}{2a}\right] = \frac{27Q^2}{80\pi\in_0 a}$

49. **A**-p, r ; **B**-q, s ; **C**-p, r ; **D**-t

(A) By inserting dielectric slab, capacitance of 1 increases there by increasing charge on capacitor 2 as more charge is flown through the battery. Energy stored in capacitor also increases.

(B) By increasing separation between the plates, capacitor C_1 decreases. Charge on C_2 also decreases

(C) By shorting capacitor 1, only capacitor 2 remains in the circuit. Potential difference across C_2 increases thereby increasing charge on 2 as well as energy stored

(D) By earthing plate of capacitor 1 potentials will change but there will be no potential difference change, making no overall change in the circuit

50.(D) For a constant current, force on rod is constant

$$F = m\frac{dv}{dt} \quad \Rightarrow \quad \int_0^v dv = \int_0^t \frac{IlB}{m}\,.\,dt$$

$$\Rightarrow \quad v = \frac{IlB}{m}\,.\,t = \frac{E_0 lBt}{mR}$$

For a constant emf E_0, the current produced by induced emf opposes the current produced by battery.

$$F = m\frac{dv}{dt} = IlB = \left(\frac{E_0 - Blv}{R}\right)lB \quad \Rightarrow \quad \frac{dv}{v - E_0/Bl} = -\left(\frac{B^2l^2}{mR}\right)dt$$

Integrating, $\int_0^v \frac{dv}{v - E_0/Bl} = -\frac{B^2l^2t}{mR}\int_0^t dt$

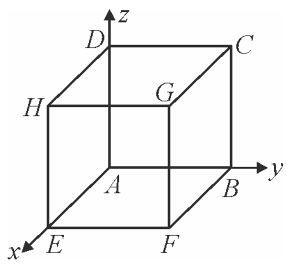
$$\Rightarrow \quad \int_0^v\left(\frac{v - E_0/Bl}{-E_0/Bl}\right) = -\frac{B^2l^2t}{mR} \quad \Rightarrow \quad v = \frac{E_0}{Bl}\left(1 - e^{-\frac{B^2l^2t}{mR}}\right)$$

With a constant current, the acceleration is a constant, so velocity does not reach terminal speed. However, with constant emf, the increasing motional emf decreases the applied force resulting in a limiting or terminal speed of $v = \frac{E_0}{Bl}$.

Hence, [I-s] [II-r] [III-p] [IV-q]

SECTION-5	NUMERICAL/SUBJECTIVE TYPE

51.(11) Flux through ABCD.



$\phi_1 = \vec{E}\vec{A} = (x^2\hat{i} + y\hat{j}).(-a^2\hat{i}) = 0$ as $x = 0$

Flux through EFGH

$\phi_2 = (x^2\hat{i} + y\hat{j}).(+a^2\hat{i}) = x^2.a^2 = a^4 = 1.0\times10^{-4}\,\text{Nm}^2/C$

Flux through BCGF

$\phi_3 = (x^2\hat{i} + y\hat{j}).(a^2\hat{j}) = a^3 = 1.0\times10^{-3}\,\text{Nm}^2/C$

Flux through EADH

$\phi_4 = (x^2\hat{i} + y\hat{j}).(-a^2\hat{j}) = 0$ as $y = 0$

Flux through ABFH

$\phi_5 = (x^2\hat{i} + y\hat{j}).(-a^2\hat{k}) = 0$

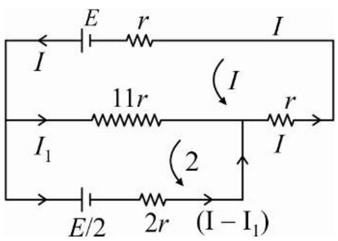
Flux through CDHG

$\phi_6 = 0$

Net flux $= (1.0\times10^{-4} + 1.0\times10^{-3})\text{N-m}^2/C = 11\times10^{-4}\,\text{N-m}^2/C$

52.(32) $\ell = 300\,cm$

$\therefore \quad r' = (275)\times\dfrac{12r}{300}$



$r' = 11r$

Using KVL in loop (1)

$E - I_1\,.\,11r - Ir - Ir = 0 \qquad ....(i)$

and in loop (2)

$-I_1 11r + (I - I_1)2r + \dfrac{E}{2} = 0 \qquad ....(ii)$

Solving equation (i) and (ii) we have

$I_1 = \dfrac{E}{16r}$ and $I = \dfrac{5E}{32r}$

So current in galvanometer

Branch $= (I - I_1) = \dfrac{5E}{32r} - \dfrac{E}{16r} = \dfrac{3E}{32r}$

$I_g = \dfrac{3E}{32r}$

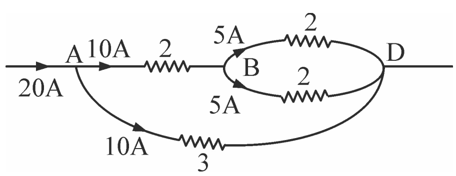
53.(300) $E < 10^6 \qquad \Rightarrow \qquad \dfrac{10^3}{d} < 10^6$

$d > 10^{-3}m^2 \qquad \Rightarrow \qquad C = \dfrac{k\varepsilon_0 A}{d}$

$d = \dfrac{k\varepsilon_0 A}{C} > 10^{-3}$

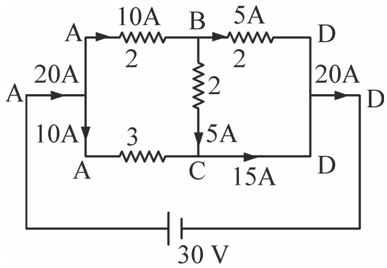
$A > \dfrac{10^{-3}\times C}{k\varepsilon_0} \qquad \Rightarrow \qquad A > \dfrac{10^{-3}\times 50\times10^{-12}}{(6\pi)\times\left(\dfrac{1}{36\pi}\times10^{-9}\right)} = 300\,mm^2$

54.(15)



$R_{eq} = \dfrac{3}{2} \qquad \Rightarrow \qquad i = \dfrac{30}{3/2} = 20$ Amp.

current distribution is as shown



55. (i) After some time t, the normal to the coil plane makes an angle ωt with the magnetic field $\vec{B} = B_0 \vec{i}$. Then, the magnetic flux through the coil is

$$\phi = N \vec{B} \cdot \vec{S}$$

where the vector surface $\vec{S}$ is given by $\vec{S} = \pi a^2 \left(cos\omega t \vec{i} + sin \;\; \omega t \vec{j} \right)$

Therefore $\qquad \phi = N \pi a^2 B_0 \;\; cos\omega t$

The induced electromotive force is

$$\varepsilon = -\frac{d\phi}{dt} \qquad \Rightarrow \qquad \boxed{\varepsilon = N \pi a^2 B_0 \omega \;\; \sin\omega t}$$

The instantaneous power is $P = \varepsilon^2 / \text{R}$, therefore

$$\boxed{\langle P \rangle = \frac{\left(N \;\; \pi \;\; a^2 B_0 \omega \right)^2}{2R}}$$

where we used $< sin^2 \omega \;\; t >= \frac{1}{T} \int_0^T sin^2 \omega t dt = \frac{1}{2}$

(ii) The total field at the center the coil at the instant t is

$$\vec{B}_t = \vec{B}_0 + \vec{B}_i$$

where $\vec{B}_i$ is the magnetic field due to the induced current $\vec{B}_i = B_i \left(\cos\omega t \vec{i} + sin \;\; \omega t \vec{j} \right)$

with $\qquad B_i = \frac{\mu_0 N I}{2a} \qquad$ and $\qquad I = \varepsilon / R$

Therefore $\qquad B_i = \frac{\mu_0 N^2 \pi a B_0 \;\; \omega}{2R} \;\; sin\omega t$

The mean values of its components are

$$\langle B_{ix} \rangle = \frac{\mu_0 N^2 \pi a B_0 \;\; \omega}{2R} \;\; \langle sin\omega t \;\; \cos \;\; \omega t \rangle = 0$$

$$\langle B_{iy} \rangle = \frac{\mu_0 N^2 \pi a B_0 \;\; \omega}{2R} \;\; \langle sin^2 \omega t \rangle = \frac{\mu_0 N^2 \pi a B_0 \;\; \omega}{4R}$$

And the mean value of the total magnetic field is

$$\langle \vec{B}_t \rangle = B_0 \;\; \vec{i} + \frac{\mu_0 N^2 \pi a B_0 \;\; \omega}{4R} \;\; \vec{j}$$

The needle orients along the mean field, therefore

$$tan\ \ \theta = \frac{\mu_0 N^2 \pi a \omega}{4R}$$

Finally, the resistance of the coil measured by this procedure, in terms of θ, is

$$\boxed{R = \frac{\mu_0 N^2 \pi a \omega}{4\,tan\ \ \theta}}$$

56. Due to cup, EF at the centre is $E = \dfrac{\sigma}{4 \in_0}$

For small angular displacement of dipole, torque on it is $\tau = PE \sin\theta \simeq PE\theta$

Restoring angular acceleration of dipole $\alpha = -\dfrac{\tau}{I} = -\dfrac{PE}{I}\cdot\theta$ .... (i)

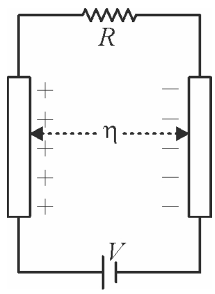
For SHM we use $\alpha = -\omega^2\theta$ .... (ii)

Comparing (i) & (ii)

$$\omega = \sqrt{\frac{PE}{I}} = \sqrt{\frac{P\sigma}{4 \in_0 I}}$$

Time period of oscillation $T = \dfrac{2\pi}{\omega} = 2x\sqrt{\dfrac{4 \in_0 I}{P\sigma}}$

57. Figure shows the situation described in the question. The two wires form a wire capacitor of which capacitance is given as $C = \dfrac{\pi\varepsilon_0 \ell}{\ell n \eta}$



Charge on the capacitor in steady state is given as

$$q = CV = \frac{\pi\varepsilon_0 \ell V}{\ell n \eta} \quad \text{. . . . . (1)}$$

Due to voltage source a constant current flows in the wires which repel each other by a magnetic force which is nullified by the force of attraction on the two wires due to opposite charges on the two wires as shown in figure. The force of attraction is given as

$$F_1 = qE = q\left(\frac{q}{2\pi\varepsilon_0 \ell(\eta r)}\right) \quad \text{. . . .(2)}$$

Substituting the value of q from equation (1) into (2),

$$F_1 = \frac{\left(\dfrac{\pi\varepsilon_0 \ell V}{\ell n \eta}\right)^2}{2\pi\varepsilon_0 \ell(\eta r)} \quad \text{. . . . . (3)}$$

The current flow is in the opposite direction in the two wires so the net force of repulsion between these wires is given as

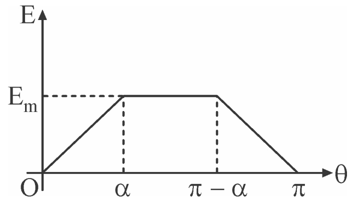
$$F_2 = \frac{\mu_0 I^2 \ell}{2\pi(\eta r)} = \frac{\mu_0 (V/R)^2 \ell}{2\pi\eta r} \quad \text{. . . . . (4)}$$

As the net force between the two wires is zero we use

$$\frac{\mu_0 (V/R)^2 \ell}{2\pi\eta r} = \frac{\left(\dfrac{\pi\varepsilon_0 \ell V^2}{\ell n \eta}\right)^2}{2\pi\varepsilon_0 \ell(\eta r)} \quad \Rightarrow \quad R = \sqrt{\frac{\mu_0}{\varepsilon_0}}\,\frac{\ell n \eta}{\pi}$$

58. (i) Output waveform is as shown

Area under curve $=\frac{1}{2}(\pi+\pi-2\alpha)E_m$



So, average value $=\frac{2(\pi-\alpha)}{2\pi}E_m=\left(\frac{\pi-\alpha}{\pi}\right)E_m$

When $\alpha=\frac{\pi}{6}$; $E_{av}=\left(\frac{\pi-\pi/6}{\pi}\right)E_m=\frac{5}{6}E_m$

When $\alpha=\frac{\pi}{2}$; $E_{av}=\left(\frac{\pi-\frac{\pi}{2}}{\pi}\right)E_m=\frac{E_m}{2}$

$\therefore$ Ratio $=\frac{\frac{5}{6}E_m}{\frac{E_m}{2}}=\frac{5}{6}\times\frac{2}{1}=\frac{5}{3}$

(ii) $E_{rms}=\sqrt{\left\{\frac{1}{\pi}\left[2\int_0^{\alpha}\frac{E_m^2}{\alpha^2}\theta^2 d\theta+\int_{\alpha}^{\pi-\alpha}E_m^2 d\theta\right]\right\}}=\left[\frac{E_m^2}{\pi}\left(2\left(\frac{\theta^3}{3\alpha^2}\right)_0^{\alpha}+(\theta)_{\alpha}^{\pi-\alpha}\right)\right]^{1/2}=E_m\sqrt{\frac{1}{\pi}\left(\pi-\frac{4\alpha}{3}\right)}$

For $\alpha=0,\ E_{rms}=E_m$

(iii) For $\alpha=\frac{\pi}{6}$; $E_{rms}=E_m\sqrt{\left(\frac{\pi-\frac{4}{3}.\frac{\pi}{6}}{\pi}\right)}=\sqrt{\frac{7}{9}}\,E_m$

For $\alpha=\frac{\pi}{2}$; $E_{rms}=\frac{E_m}{\sqrt{3}}$ $\quad\therefore$ Ratio $=\sqrt{\left(\frac{7}{9}\times\frac{3}{1}\right)}=\sqrt{\frac{7}{3}}$

59. (i) The total charge Q is just the area under the curve of the figure. Because of the triangular shape, we immediately get

$$Q=\frac{I_0\tau}{2}=5C.$$

(ii) The average current is

$$I=Q/\tau=\frac{I_0}{2}=50kA,$$

Simply half the maximal value.

(iii) Since the bottom of the cloud gets negatively charged and the ground positively charged, the situation is essentially that of a giant parallel-plate capacitor. The amount of energy stored just before lightning occurs is $QE_0h/2$ where E_0h is the voltage difference between the bottom of the cloud and the ground, and lightning releases this energy. Altogether we thus get for one lightning the energy $QE_0h/2=7.5\times10^8 J$. It follows that you could light up the 100W bulb for the duration

$$t=\frac{32\times10^6}{6.5\times10^9}\times\frac{7.5\times10^8 J}{100W}\approx10h.$$

60. (i) (a) For the given vlaue of x, the amount of charge on each capacitor is

$$Q_1 = VC_1 = \frac{\varepsilon_0 AV}{d-x}$$

$$Q_2 = VC_2 = \frac{\varepsilon_0 AV}{d+x}$$

(b) Note that we have two capacitors. By using the formula for force between plates of a capacitor, we get

$$F_1 = \frac{Q_1^2}{2\varepsilon_0 A}, \qquad F_2 = \frac{Q_2^2}{2\varepsilon_0 A}$$

As these two forces are in the opposite directions, the net electric force is

$$F_E = F_1 - F_2 \quad \Rightarrow \quad F_E = \frac{\varepsilon_0 AV^2}{2}\left(\frac{1}{(d-x)^2} - \frac{1}{(d+x)^2}\right)$$

Ignoring terms of order x^2 in the answer we get

$$F_E = \frac{2\varepsilon_0 AV^2}{d^3}x$$

(c) There are two springs placed in series with the same spring constant, k, then the mechanical force is

$$F_m = -2kx$$

Combining this result and noticing that these two forces are in the opposite directions, we get

$$F = F_m + F_E \Rightarrow \quad F = -2\left(k - \frac{\varepsilon_0 AV^2}{d^3}\right)x,$$

By using the Newton's second law,

$$F = ma$$

And the answer, we get

$$a = -\frac{2}{m}\left(k - \frac{\varepsilon_0 AV^2}{d^3}\right)d$$

(ii) Starting with Kirchhoff's laws, for two electric circuits, we have

$$\begin{cases} \dfrac{Q_s}{C_s} + V - \dfrac{Q_2}{C_2} = 0 \\ -\dfrac{Q_s}{C_s} + V - \dfrac{Q_1}{C_1} = 0 \\ Q_2 - Q_1 + Q_s = 0 \end{cases}$$

Noting that $V_s = \frac{Q_s}{C_s}$ one obtains $\Rightarrow$ $V_s = \dfrac{\dfrac{2\varepsilon_0 Ax}{d^2 - x^2}}{C_s + \dfrac{2\varepsilon_0 Ad}{d^2 - x^2}}$.

Ignoring terms of order x^2 in the answer we get

$$V_s = V\frac{2\varepsilon_0 Ax}{d^2 C_s + 2\varepsilon_0 Ad}$$